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599 Edge AI and Robotics
Module 5 OpenAI Gym

Objective:

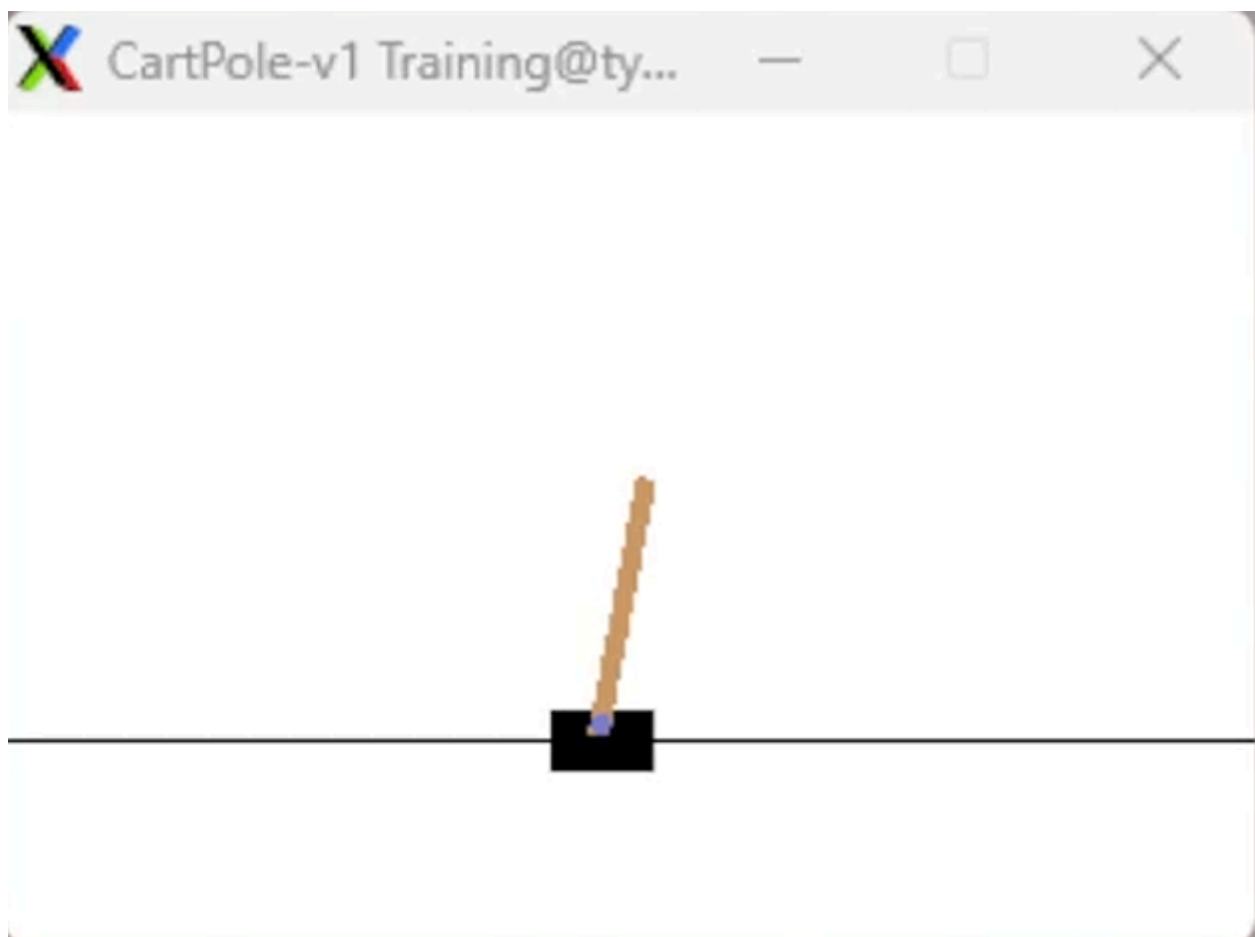
The objective of this lab was to implement and observe Reinforcement Learning (RL) agents using the OpenAI Gym. The lab provides a pre configured Docker container for the Jetson platform and demonstrated how autonomous agents learn through trial and error by interacting with environments such as CartPole and Lunar Lander.

Environment:

The setup utilized the jetson-reinforcement repository. The use of Docker simplified the installation of dependencies such as PyTorch, OpenAI Gym, and NVIDIA graphics drivers for the Jetson Nano.

CartPole-v1:

The CartPole environment is a problem where the agent is tasked with balancing a vertical pole on a moving cart. The agent receives a reward for every timestep the pole remains upright.



Observation:

Using the Advantage Actor-Critic (A2C) algorithm, the agent quickly improved its performance. The test is considered complete when the agent reaches a maximum score of 200 timesteps with the pole remaining balanced. A video of the evaluation phase is available under media.

```
episode 192 steps 14811 ep_reward 500.00 ep_reward_mean 114.08
Saving best model with reward=500.00 to /jetson-reinforcement/data/models/CartPo
le-v1_A2C_state.zip
Stopping training because the last episode reward 500.00 is above the threshold
475.0
```

Lunar Lander-v2:

The Lunar Lander environment is a more complex challenge that involves gravity and momentum. The agent's goal is to navigate the lander to a designated landing pad between two flags.

**Observation:**

Points are awarded based on the proximity to the landing pad, fuel consumption, and the stability of the landing gear touching down. Even in this more complex setup the agent performs quite well given enough time. A video of the evaluation phase is available under media.

```
episode 51  steps 9753  ep_reward -22.25  ep_reward_mean -216.02
```

rollout/		
ep_len_mean	188	
ep_rew_mean	-216	
time/		
fps	34	
iterations	2000	
time_elapsed	292	
total_timesteps	10000	
train/		
entropy_loss	-0.532	
explained_variance	-38.3	
learning_rate	0.0007	
n_updates	1999	
policy_loss	0.696	
value_loss	11.6	

Conclusion:

This lab demonstrated the integration of the OpenAI Gym toolkit with NVIDIA's provided containers. By observing the CartPole and Lunar Lander simulations, the core concepts of states, actions, and reward signals in reinforcement learning were able to be seen in real time.